A

Run	Project · material · age (years)	C1	C2	C3	C4	C5	Reusable	PaleoRigor	What did not match
ERR2269804	Bieker2020 · leaf · 100	✓	✓	✓	✓	✓	✓	executed	
ERR4172835	Brealey2020 · dental calculus · 100	✓	✓	✓	✓	✓	✓	executed	
ERR6097630	Brealey2021 · dental calculus · 100	✓	✓	✓	✓	✗	✗	executed	ENA count = both mates
SRR6933820	McDonough2018 · keratinous claw · 100	✓	✓	✓	✓	✓	✓	executed	
ERR3589349	Velsko2019 · dental calculus · 200	✓	✓	✓	✓	✓	✓	executed	
ERR6589623	Kazarina2021b · tooth · 400	✓	✓	✓	✓	✓	✓	executed	
SRR1047998	Appelt2014 · palaeofaeces · 600	✗	✓	✓	✓	✓	✗	executed	catalogue entry differs from ENA
SRR2229975	Ozga2016 · tooth · 700	✓	✓	✓	✗	–	✗	**blocked**	one file, listed paired-end
SRR13355810	Farrer2021 · dental calculus · 800	✓	✓	✓	✗	–	✗	**blocked**	one file, listed paired-end
SRR5047044	Philips2017 · tooth · 1,000	✓	✓	✓	✓	✓	✓	executed	
ERR11660455	Jackson2024 · tooth · 3,800	✓	✓	✓	✓	✓	✓	executed	
ERR4863268	SeguinOrlando2021 · bone · 4,400	✓	✓	✓	✓	✗	✗	executed	ENA count = both mates
ERR1094779	Maixner2016 · digestive tract contents · 5,300	✓	✓	✓	✓	✓	✓	executed	
SRR17604766	Quagliariello2022 · dental calculus · 6,200	✓	✓	✓	✗	✗	✗	executed	extra unpaired-read file
SRR13263133	Weyrich2017 · dental calculus · 7,400	✓	✓	✓	✓	✓	✓	executed	
ERR3579720	FellowsYates2021 · dental calculus · 31,000	✓	✓	✓	✓	✓	✓	executed	
ERR14100444	Jin2025 · digestive tract contents · 52,500	✓	✓	✓	✓	✗	✗	executed	ENA count = both mates

✓ passed ✗ failed – not evaluated (no PaleoRigor output). blocked: refused before execution (missing_mate). Records ordered by sample age.
C1 catalogue files and sizes = ENA C2 catalogue read count = ENA C3 downloaded MD5 = ENA C4 layout consistent C5 FastQC reads per mate = ENA read count

B

Records reusable exactly as catalogued

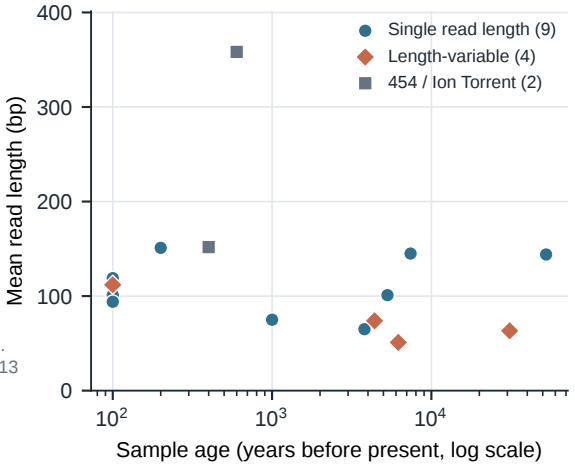
10 of 17 (58.8%; Wilson 95% CI, 36.0–78.4%)



- Reusable as catalogued (10)
- ENA read count covers both mates (3)
- one file for a paired-end library (2)
- extra unpaired-read file (1)
- catalogue file entry differs from ENA (1)

PaleoRigor decision matched the layout check in 17 of 17 records. Reads were length-variable (processed before deposition) in 4 of 13 assessable Illumina records.

C



Descriptive only: one library per project; executed records only.